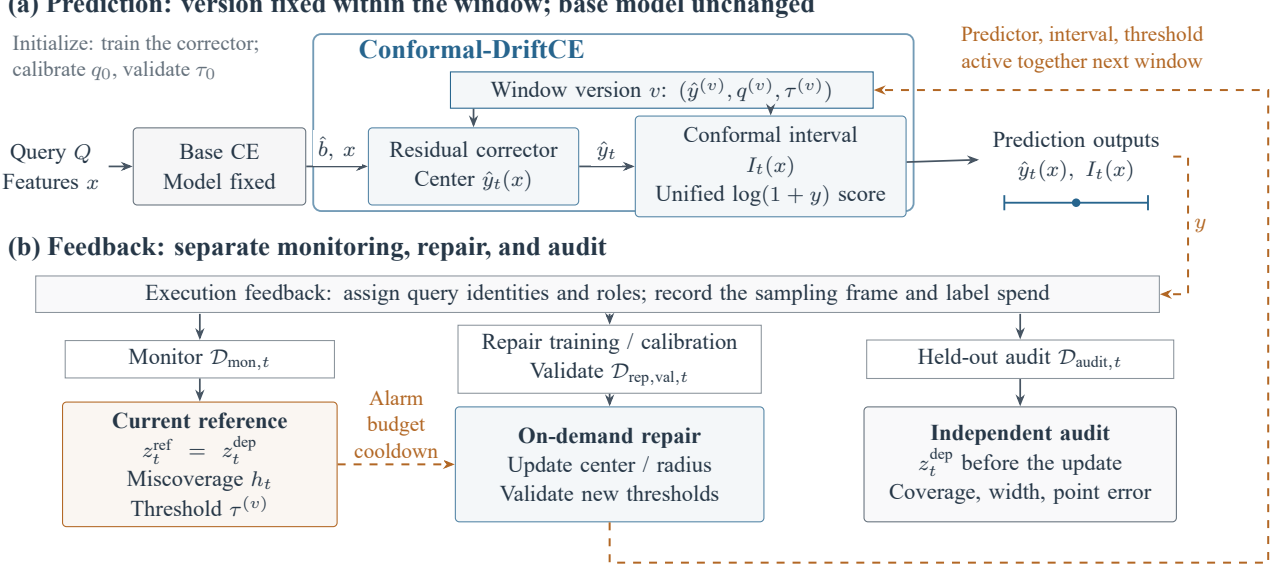

(a) Prediction: version fixed within the window; base model unchanged
Initialize: train the corrector; calibrate q_0, validate τ_0
Conformal-DriftCE
Predictor, interval, threshold active together next window
Window version v: $(\hat{y}^{(v)}, q^{(v)}, \tau^{(v)})$
Query Q
Features x
Base CE Model fixed
$\hat{b}$, x
Residual corrector Center $\hat{y}_t(x)$
$\hat{y}_t$
Conformal interval $I_t(x)$ Unified $\log(1+y)$ score
Prediction outputs $\hat{y}_t(x)$, $I_t(x)$
y
(b) Feedback: separate monitoring, repair, and audit
Execution feedback: assign query identities and roles; record the sampling frame and label spend
Monitor $\mathcal{D}_{\text{mon},t}$
Repair training / calibration Validate $\mathcal{D}_{\text{rep,val},t}$
Held-out audit $\mathcal{D}_{\text{audit},t}$
Current reference $z_t^{\text{ref}} = z_t^{\text{dep}}$ Miscoverage h_t Threshold $\tau^{(v)}$
Alarm budget cooldown
On-demand repair Update center / radius Validate new thresholds
Independent audit z_t^{dep} before the update Coverage, width, point error
Solid: predictions or labels
Dashed: feedback control / delayed updates
Audit never feeds back to training, calibration or threshold selection
Fixed-initial-reference control always uses $(\hat{y}_0, q_0, \tau_0)$; the main path uses the current reference consistent with the deployed version.